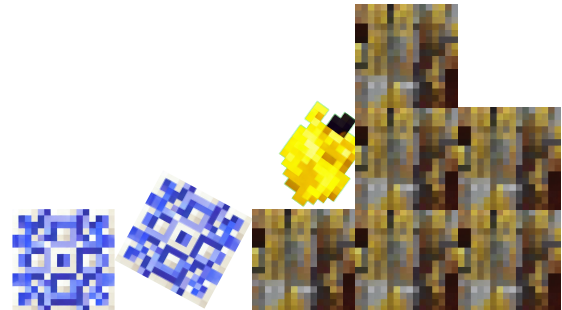


M¹nettexture

INFO9023 - Milestone 1

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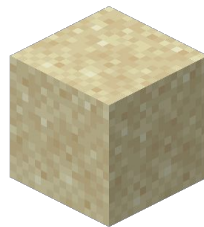
Reminder

- Minecraft: Sandbox game in a world made of blocks
 - Lots of texture to personalize
- Project goal: Generate blocks' texture from a textual prompt

In game screenshot



Sand block



Sand texture (1 side)



Sand block texture in different resource packs



[Halloween mash-up](#)

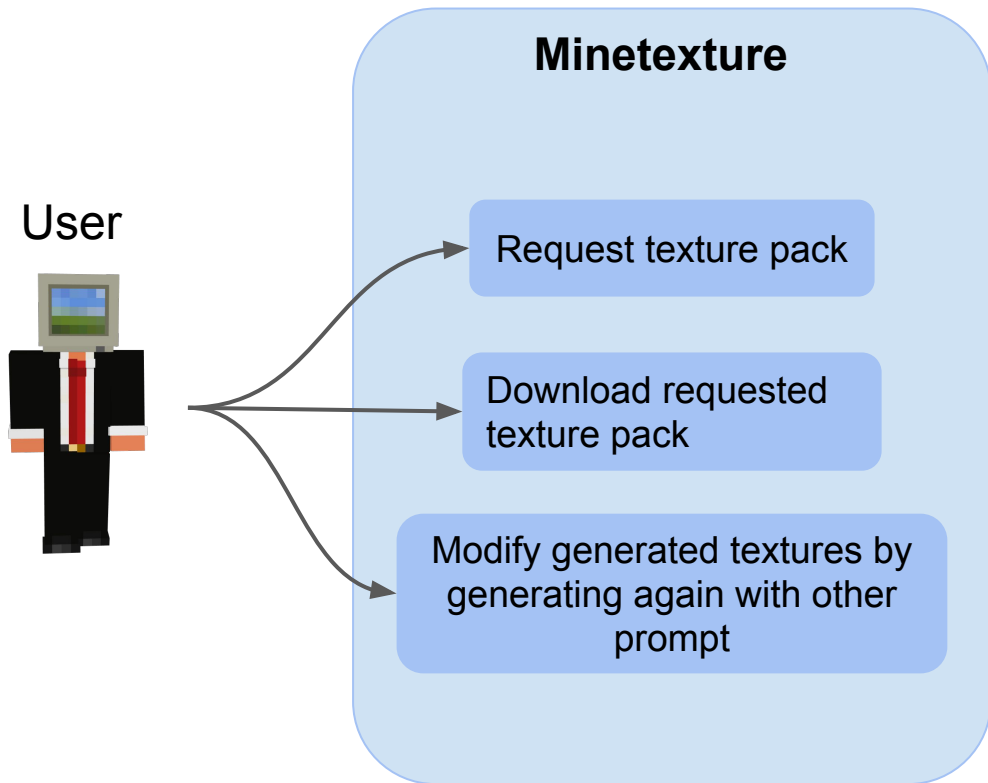


[Super-cute](#)



[Jolicraft](#)

Use case



For:

- **Developers** that need quick mod iterations
- **Texture pack makers** to get ideas for new textures
- **Players** that wants to customize their texture pack

It is **NOT** a replacement for real artists, instead helps on fast iterations

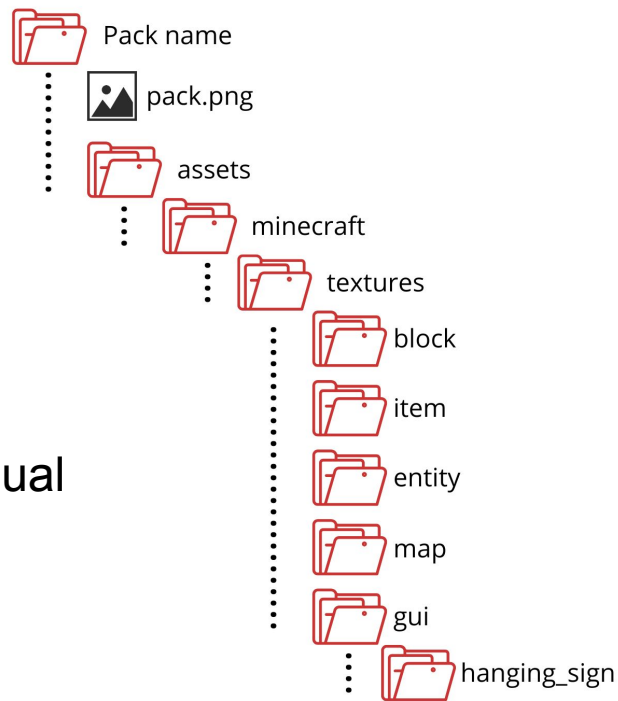
Training Data - Overall structure of a texture pack

A ***minecraft texture pack*** is a collection of files that modifies the visual appearance of the game.

- Textures
- Sounds
- Shaders
- Configuration files
- ...

→ Only focus on image files that represent visual objects

Style.json with : style, keywords



Training Data - MineTexture dataset

Our dataset:

Style learning 

- 10 Texture packs
- Redesign existing objects
- Provide stylistic diversity

21 788 images

Content learning 

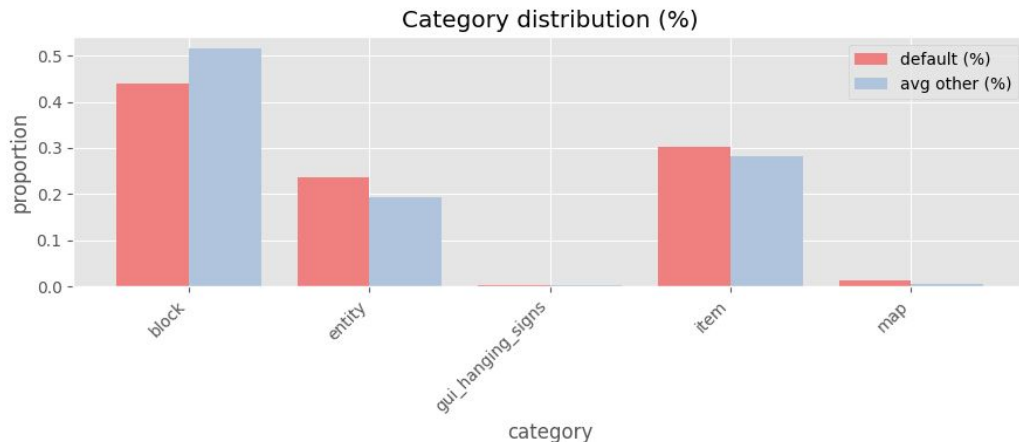
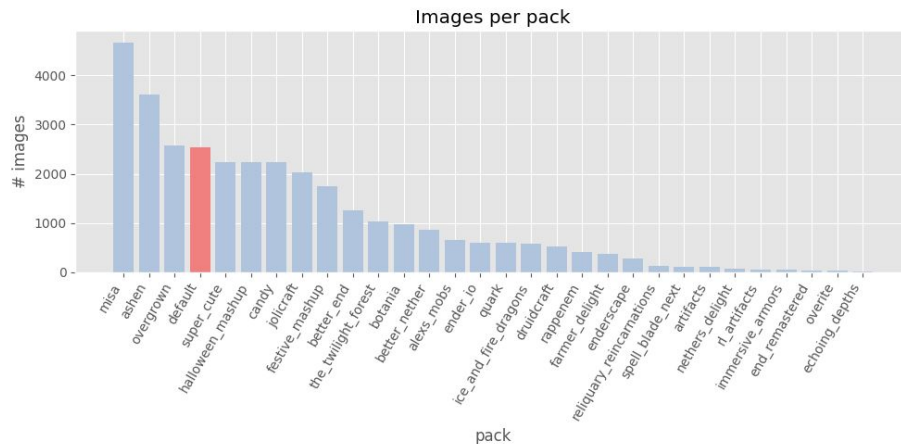
- 19 Mods
- Add new objects
- Expand the object vocabulary

8 295 images

—————> 30 083 texture images in total

Training Data - MineTexture dataset

- Large variability across packs
- Some packs dominate over default pack
- A lot of packs have less than 500 images



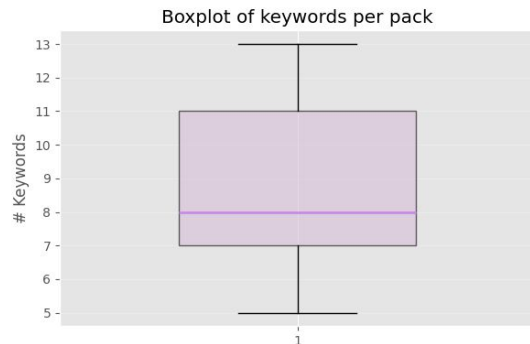
- Similar distribution across categories
- No extreme imbalance

→ The dataset is structurally consistent

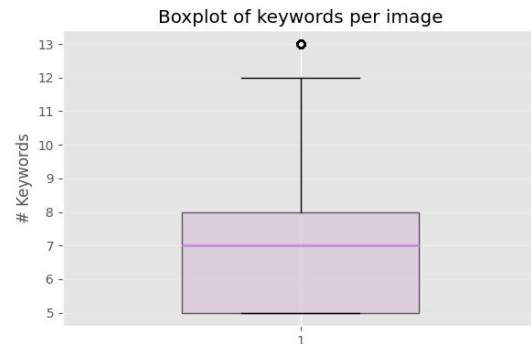
Training Data - MineTexture dataset

Total distinct number of keywords: 190

Colorful	6909
Fantasy	6563
Playful	5326
Realistic	5086
Medieval	5253
Detailed	5086
Cute	4814
Natural color	4670
Subtle shading	4670
Soft lighting	4670



- Median = 8 keywords per images
- Between 5 and 13 keywords



- Median = 7 keywords per images
- Between 5 and 8 keywords

Training Data - Image preprocessing pipeline

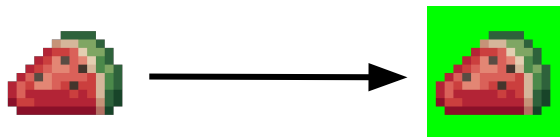
- Resize

- To 512x512
- Method: Nearest Neighbor interpolation

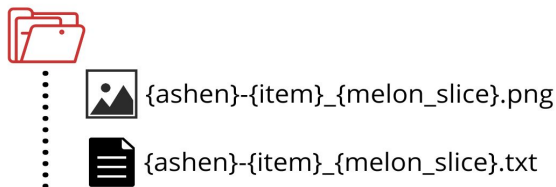


- Background processing

- GREEN: (0, 255, 0, 255)



- Final structure



Model experiments

- Experiments on Quadro RTX 6000 (24GB)
- 3 stable diffusion models
 - Stable Diffusion XL (13s/request for 1024x1024 resolution)



Image Generated by **SDXL**

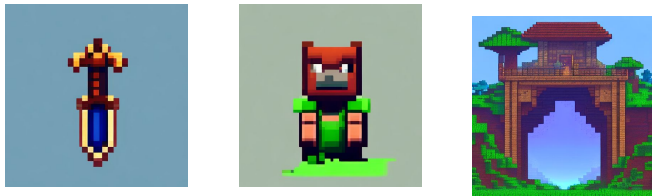
Prompt: "minecraft texture pixel art, item, diamond sword, theme medieval dark steampunk industrial"

- Stable Diffusion 2.1 (10s/image for 256x256 resolution)
- Stable Diffusion 1.5 (2-4s/image for 512x512 resolution)

Model experiments

- 3 fine-tuned models

- All-in-one: DreamBooth fine-tuned model, generic pixel art generation (SD 1.5)



- Minecraft Texture: LoRA fine-tuned model, minecraft specific item texture generation (SD 2.1)



- Poxel Minecraft: LoRA fine-tuned model, minecraft specific item texture generation (SD 1.5)



Model training - System Design

- Containerized training environment for reproducibility (Docker)
- Training parameters, paths, WandB API key etc... pulled from the environment (.env file)
- Timestamped training runs
- Python environment loaded from UV project (pyproject.toml)
- Training data fetched from Google Cloud Storage in bucket training_data_mintexture
- Training supervision using WandB



Accelerate



Dffusers



PyTorch



Weights & Biases

by CoreWeave

Model training - LoRA fine-tuning

- Base model: Stable Diffusion 1.5
- Small training subset: 1000 images
- GPUs: 2 x Quadro RTX 6000 (2 x 24GB)
- Training parameters:
 - LoRA parameters: Rank = 32, Alpha = 16
 - Max token length: 150, caption token dropout: 0.05
 - Batch size: 8, Epochs: 25, No mixed precision
 - Learning Rate (UNet) = $1e-4$, (Text Encoder) = $1e-4$, AdamW



Model training - Results

Prompt:

ancient will ahrim, item,...

twig magical wand bind, item,...

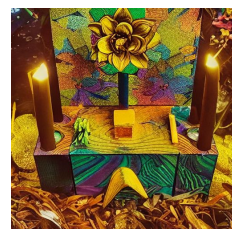
yellow shiny flower, block, magical,...

yellow shiny heart, block, magical,...

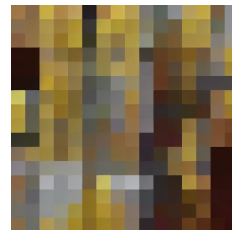
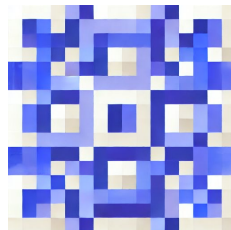
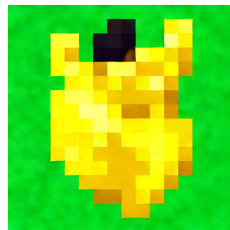
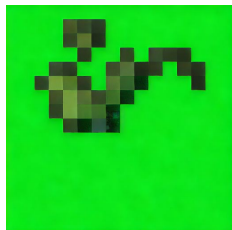
azulejo 1, block, magical,...

yellow shiny wood,...

SD 1.5:



Ours:



The green background can be removed with the [GrabCut algorithm](#) available in OpenCV (150ms / image)

Questions ?

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